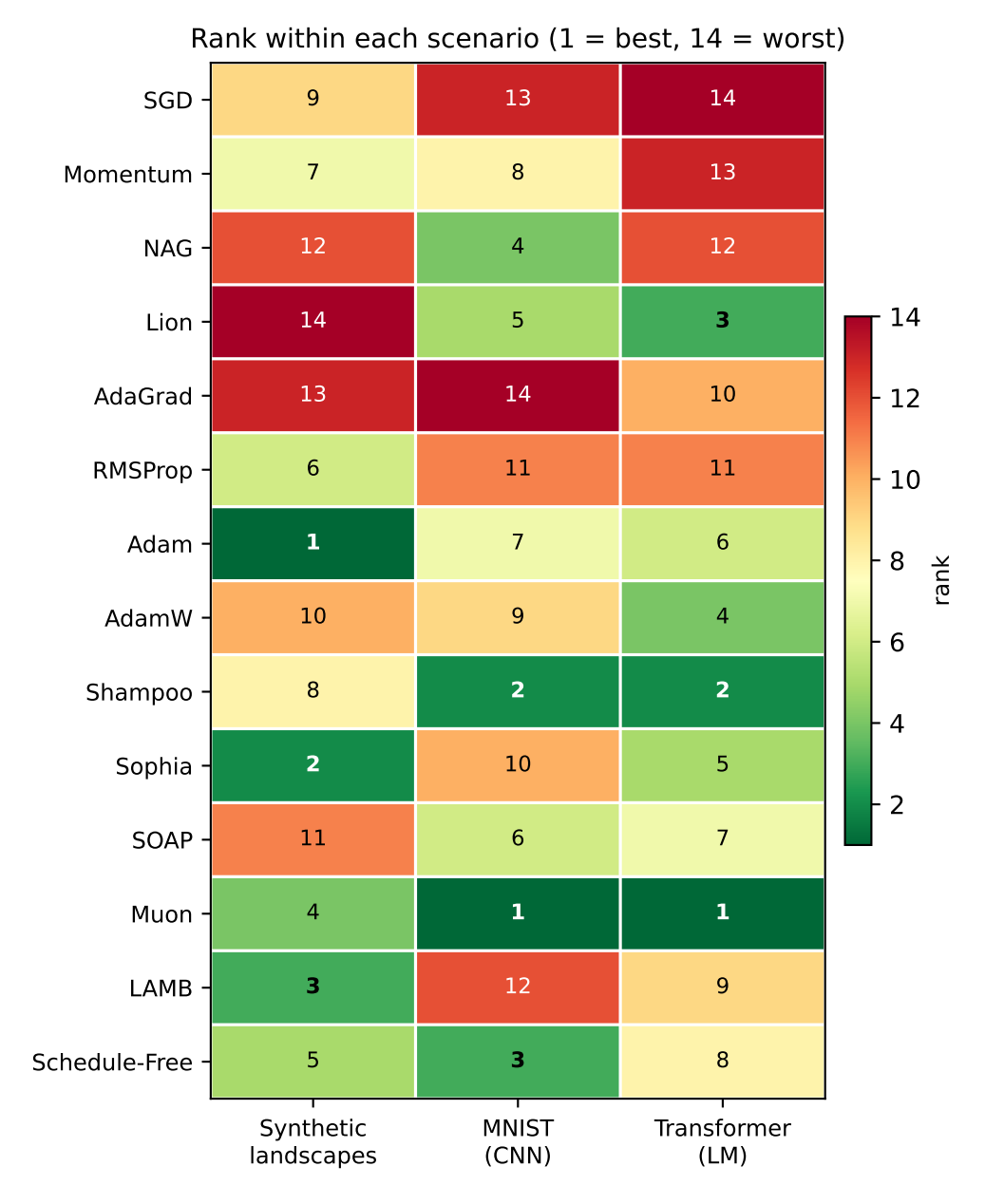
Rank within each scenario (1 = best, 14 = worst)
SGD
Momentum
NAG
Lion
AdaGrad
RMSProp
Adam
AdamW
Shampoo
Sophia
SOAP
Muon
LAMB
Schedule-Free
9 13 14
7 8 13
12 4 12
14 5 3
13 14 10
6 11 11
1 7 6
10 9 4
8 2 2
2 10 5
11 6 7
4 1 1
3 12 9
5 3 8
Synthetic landscapes
MNIST (CNN)
Transformer (LM)
rank
14
12
10
8
6
4
2